

1

STRUCTURAL DRAWINGS TO BE READ IN CONJUNCTION WITH STRUCTURAL SPECIFICATIONS. ALL ARCHITECTS AND SERVICES DRAWINGS AND SPECIFICATIONS.

2

THE CONTRACT STRUCTURAL DRAWINGS AND SPECIFICATIONS REPRESENT THE FINISHED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION. THE CONTRACTOR SHALL PROVIDE ALL MEASURES NECESSARY TO PROTECT THE STRUCTURE DURING CONSTRUCTION. SUCH MEASURES SHALL INCLUDE, BUT NOT BE LIMITED TO BRACING, SHORING OF LOADS DUE TO CONSTRUCTION EQUIPMENT, ETC. BEFORE RELATED WORK COMMENCES THE CONTRACTOR SHALL SUBMIT A METHOD STATEMENT AND SEQUENCE OF WORK TO THE ENGINEER AND ARCHITECT.

3

THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND ELEVATIONS WITH ARCHITECTURAL DRAWINGS PRIOR TO START OF CONSTRUCTION. ALL DISCREPANCIES TO BE NOTIFIED, IN WRITING TO ENGINEERS AND ARCHITECTS FOR RESOLUTION. ALL SETTING-OUT TO BE TAKEN FROM THE ARCHITECTS DRAWINGS.

4

CONSTRUCTION MATERIAL SHALL BE SPREAD OUT IF PLACED ON FRAMED FLOORS OR ROOF, IN SUCH A MANNER THAT THE STRUCTURE IS NOT OVERLOADED IN EITHER THE PERMANENT OR TEMPORARY CONDITION? WHERE REFERENCE IS MADE TO VARIOUS TEST STANDARDS FOR MATERIALS, SUCH STANDARDS SHALL BE THE LATEST EDITION AND/OR ADDENDUM. OTHER STANDARDS MAY BE CONSIDERED BUT ONLY WITH PRIOR APPROVAL BY THE ENGINEER.

5

CONTRACTOR TO ESTABLISH AND VERIFY ALL OPENINGS AND INSERTS FOR ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING WITH APPROPRIATE TRADES, DRAWINGS AND SUBCONTRACTOR PRIOR TO CONSTRUCTION.

6

OPTIONS ARE FOR CONTRACTORS CONVENIENCE. HE SHALL BE RESPONSIBLE FOR ALL CHANGES NECESSARY IF HE CHOOSES AN OPTION AND HE SHALL COORDINATE ALL DETAILS.

7

ALL DIMENSIONS ON DRAWINGS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED. WRITTEN DIMENSIONS ONLY APPLY. DRAWINGS ARE NOT TO BE SCALED.

8

FOR CLARITY, ALL ROOF AND FLOOR OPENINGS MAY NOT BE SHOWN ON FRAMING PLANS. FOR EXACT SIZE, NUMBER AND LOCATION FOR OPENING, SEE ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS. WHERE OPENINGS ARE INDICATED ON THE STRUCTURAL DRAWINGS, THESE ARE TO BE CHECKED BY THE CONTRACTOR AGAINST RELEVANT SERVICE ENGINEERS OR SUBCONTRACTORS DRAWINGS PRIOR TO CONSTRUCTION.

9

ALL LEVELS ARE RELATED TO THE ORDNNANCE DATUM (OD)

10

FOR LOCATION AND DETAILS OF ELECTRICAL EARTHING OF STRUCTURE REFER TO ELECTRICAL DRAWINGS AND DETAILS.

11

FOR FIRE PROOFING REQUIREMENTS REFER TO RELEVANT ARCHITECTS DRAWINGS AND RELEVANT FIRE SAFETY CERTIFICATE DRAWINGS.

12

THE CONTRACTOR SHALL NOTE, AND MAKE ALLOWANCES FOR, THE MEASURES NECESSARY TO COMPLY WITH THE WASTE MINIMIZATION AND RECYCLING TARGETS SET OUT IN THE SPECIFICATION.

13

ALL DPMs, DPCs, RADON PROTECTION BARRIERS AND SUMPS, ETC. TO ARCHITECTS SPECIFICATION & DETAILS.

FOUNDATIONS

1

PRELIMINARY TRIAL HOLES CARRIED OUT: NO

IF YES, APPROX. LOCATION OF TRIAL HOLES SHOWN ON DWg No. 0000

2

FURTHER SITE INVESTIGATION TO BE CARRIED OUT BY: INDEPENDENT LABORATORY: INDEPENDENT LABORATORY DATE: DATE:

3

FOR LEVELS TO TOP OF ALL FOUNDATIONS REFER TO FOUNDATION LAYOUT DRAWING & SECTIONS THEREOF

4

CONCRETE BLINDING SHALL BE PROVIDED UNDER ALL FOUNDATIONS TO A MINIMUM THICKNESS OF 75mm. EXCAVATED SURFACES SHALL BE FREE OF LOOSE MATERIAL, DRY AND BLINDED AS SOON AS POSSIBLE AFTER INSPECTION OF BEARING SURFACES BY THE ENGINEER.

5

FOUNDATIONS TO BE CENTERED UNDER COLUMNS AND WALLS UNLESS NOTED OTHERWISE ON THE DRAWINGS.

1

ALL PREFABRICATED TIMBER TRUSSED RAFTERS SHALL BE MANUFACTURED, SUPPLIED AND ERECTED IN ACCORDANCE WITH IS EN 1995-1-1 DESIGN OF TIMBER STRUCTURES AND IS EN 14250.

2

ALL PREFABRICATED TIMBER TRUSSED RAFTERS SHALL BE FIXED TO THE WALL PLATES USING "BAT" TRUSS CLIPS, OR EQUAL AND APPROVED, DESIGNED AND SUPPLIED BY THE TRUSSED RAFTER MANUFACTURER. ALL TRUSS CLIPS TO BE FIXED TO WALL PLATE AND TRUSSES USING 32mm X 9 GAUGE SQUARE TWISTED SHERADISED NAILS THROUGH DUAL HOLES IN THE TRUSS CLIPS. MULTIPLE TRUSS GIRDERS SHALL BE FIXED TO THE WALL PLATE USING CONNECTORS DESIGNED BY THE TRUSSED RAFTER MANUFACTURER TO RESIST THE APPROPRIATE DESIGN UPLIFT LOADS.

3

ALL PREFABRICATED TIMBER TRUSSED RAFTERS SHALL HAVE VACUUM PRESSURE IMPREGNATED PRESERVATIVE TREATMENT OF THE ORGANIC SOLVENT TYPE APPLIED. THE SOLVENT MUST BE COMPATIBLE WITH METAL PLATE FASTENERS, TRUSS CLIPS ETC., AND MUST NOT CAUSE CORROSION.

4

DUMMY TILING BATTENS SHALL BE INSTALLED @ 600C/C BETWEEN REDUCING TRUSSES.

5

ALL 25 X 100 DIAGONAL AND LONGITUDINAL BRACING AND BINDERS ARE TO BE NAILED WITH TWO 3.35mm DIAMETER X 75mm LONG GALVANIZED ROUND WIRE NAILS TO EVERY TRUSSED RAFTER THEY CROSS.

6

LONG BRACES AND BINDERS MAY BE LAP JOINTED PROVIDED THE OVERLAP IS NAILED TO AT LEAST TWO TRUSSED RAFTERS.

7

THE LONGITUDINAL AND TRANSVERSE BINDERS SHALL ABUT GABLE AND PARTY WALLS.

8

ALL BRACING TO CEILING TIES, RAFTER MEMBERS AND INTERNAL MEMBERS OF TRUSSED RAFTERS SHALL BE ERECTED AND THEN APPROVED BY THE ENGINEER BEFORE FIXING OF SLATING BATTENS.

9

THE QUALITY OF TIMBER TO BE USED FOR THE BRACING MEMBERS (COMMON AND CEILING JOIST, IF ANY), SHALL BE IMPORTED EUROPEAN WHITEWOOD GS GRADE OR HOMEGROWN SITKA SPRUCE SS GRADE IN ACCORDANCE WITH IS EN 14081-1 AND IS EN1912, DRIED TO 20% MAXIMUM MOISTURE CONTENT TO THE FINISHED SIZES SPECIFIED ON THE DRAWING. A CERTIFICATE OF COMPLIANCE WITH REGARD TO MOISTURE CONTENT AND GRADE STRESS MUST BE SUPPLIED BY THE MAIN CONTRACTOR.

10

ALL BRACING MEMBERS, VALLEY BOARDS, RIDGE AND HIP BOARDS SHALL HAVE BEENPRETREATED WITH AN ORGANIC SOLVENT TYPE PRESERVATIVE UPON DELIVERY TO SITE.

11

SLATING BATTENS SHALL BE 50 X 35 @ 250C/C FIXED BY 10 GAUGE GALVANIZED WIRE NAILS AT LEAST 32MM LONGER THAN THE BATTEN THICKNESS. BATTENS SHALL BE FIXED AT EACH RAFTER WITH AT LEAST ONE NAIL. NO MORE THAN ONE BATTEN IN FOUR SHALL BE JOINED OVER EACH RAFTER.

12

THE MAIN CONTRACTOR SHALL MAKE ADEQUATE PROVISION FOR DELIVERY, OFF LOADING AND STORAGE OF PREFABRICATED TRUSSED RAFTERS ON SITE. THE TRUSSES SHALL BE STORED VERTICALLY ON RAISED BEARERS (SUPPLIED BY THE MAIN CONTRACTOR) TO AVOID CONTACT WITH THE GROUND. THE BEARERS SHALL BE LOCATED AT THE POINTS OF SUPPORT ASSUMED IN THE DESIGN AND AT SUCH A HEIGHT AS TO ENSURE THAT THE RAFTER OVERHANGS CLEAR THE GROUND. THE MAIN CONTRACTOR'S ATTENTION IS DRAWN IN PARTICULAR TO BS8000 REGARDING THE HANDLING OF MATERIALS AND INFORMATION TO BE SUPPLIED BY THE TRUSS MANUFACTURER.

13

WHERE CEILING JOISTS AND MONO TRUSSES ARE SUPPORTED ON TRUSS GIRDERS, THE CONNECTIONS SHALL BE MADE USING JOIST HANGERS OF ADEQUATE DESIGN STRENGTH.

14

WALL PLATES SHALL BE FIXED TO SOLID BLOCKWORK & RC BEAMS BY USING PROPRIETARY GALVANIZED "L" SHAPED HOLDING DOWN STRAPS AT 1500C/C. THE STRAPS SHALL BE OF APPROVED MANUFACTURE WITH A MINIMUM GROSS SECTION OF 30X5MM & OVERALL LENGTH OF 1000MM WITH A LONG LEG OF 850MM. ALTERNATIVELY IN THE CASE OF RC BEAMS THE WALL PLATES MAYBE FIXED TO THE BEAMS WITH 12MM DIA HOLDING DOWN BOLTS CAST IN AT 1000MM C/C (WASHER UNDER HEAD). FINAL DETAILS FOR FIXING WALL PLATES ARE TO BE APPROVED BY THE ARCHITECT.

1

ALL STRUCTURAL STEELWORK SHALL BE IN ACCORDANCE WITH BS5950 AND BSEN 10025 OR BSEN 10210 PART 1. GRADE S275 SHALL BE PROVIDED EXCEPT WHERE NOTED OTHERWISE.

2

STEEL GRADES UNLESS NOTED OTHERWISE ON DRAWINGS SHALL BE:

*UB /UC: S355 TO BS EN 10025
*CHANNELS/PLATES: S355 TO BS EN 10025
*CHS/SHS/RSH: S355 TO BS EN 10210
*EQUAL/UNEQUAL RSA: S355 TO BS EN 10056
*KICK FLATS/BARS: S355 TO BS EN 10025

S 275 JR FOR MATERIAL OF THICKNESS ≤20mm.
S 275 JO FOR MATERIAL OF THICKNESS >20mm, ≤40mm.
S 275 J2G3 FOR MATERIAL OF THICKNESS >40mm, <90mm.
S 275 JOH FOR HOLLOW SECTIONS THICKNESS ≤ 40mm.

WHERE GRADE S 355 IS SPECIFIED, THE PARTICULAR QUALITY GRADE FOR EACH ELEMENT SHALL BE AS FOLLOWS:-
S 355 JR FOR MATERIAL OF THICKNESS ≤ 25mm.
S 355 JO FOR MATERIAL OF THICKNESS > 25mm, ≤ 45mm.
S 355 J2G3 FOR MATERIAL OF THICKNESS > 45mm, ≤ 100mm.
S 355 JOH FOR ROLLED HOLLOW SECTIONS OF THICKNESS ≤ 45mm.

3

ALL STEEL SHALL BE SAWN CUT.

4

ANCHOR BOLTS SHALL BE GRADE 8.8 MATERIAL UNLESS STATED OTHERWISE ON THE DRAWINGS. GROUT UNDER BASE PLATES SHALL BE SBD FIVE STAR NON SHRINK GROUT MIXED AND INSTALLED IN STRICT ACCORDANCE WITH THE MANUFACTURERS RECOMMENDATIONS. HOLDING DOWN BOLTS AND LEVELING PACKS SHALL BE TOTALLY ENCLOSED BY GROUT.

5

BLAST CLEAN STEEL TO SA 2 1/2. FINISH TO BE PAINTED OR GALVANIZED AS INDICATED ON THE ARCHITECTURAL G.A. DRAWINGS OR SPECIFICATIONS.

6

COLD FORMED PURLINS AND SHEETING RAILS SHALL BE DESIGNED TO IS EN1993 AND SHALL BE MANUFACTURED FROM HOT DIPPED GALVANIZED STEEL TO IS EN 10147: 992. STEEL SHALL BE GRADE E350 WITH A Z225 ZINC COATING UNLESS NOTED OTHERWISE.

7

ALL CONNECTIONS NOT SHOWN ON THE ENGINEERS DRAWINGS SHALL BE DESIGNED AND DETAILED BY THE CONTRACTOR. CALCULATIONS SHALL BE SUBMITTED FOR APPROVAL WITH THE FABRICATION DRAWINGS.

8

THE CONTRACTOR SHALL SUBMIT FULLY DETAILED FABRICATION DRAWINGS TO THE ENGINEER FOR APPROVAL A MINIMUM OF 10 WORKING DAYS BEFORE FABRICATION IS DUE TO COMMENCE. NOTARIZATION SHALL COMMENCE UNTIL APPROVAL OF THE SHOP DRAWINGS IS RECEIVED AND UNTIL ALL COMMENTS HAVE BEEN INCORPORATED.

9

APPROVAL BY THE ENGINEER IN NO WAY RELIEVES THE CONTRACTOR FOR ANY RESPONSIBILITY FOR THE ACCURACY, CORRECTNESS AND ADEQUACY OF CALCULATIONS, DESIGN, DETAILS AND DIMENSIONS.

10

ALL STEELWORK BELOW GROUND TO BE ENCASED IN CONCRETE. MINIMUM COVER TO STEEL TO BE 75mm.

11

ALL CONTACT SURFACES IN FRICTION GRIP BOLTED CONNECTIONS TO BE LEFT UNPAINTED.

12

ANY EXPOSED LOAD BEARING STEELWORK TO BE PAINTED WITH INTUMESCENT PAINT TO PROVIDE 1 Hr FIRE RATING REFER TO FIRE CERT DRAWINGS.

13

ALL COLUMNS, RAFTERS AND LOAD CARRYING BEAMS AND CONNECTIONS TO BE 1hr FIRE PROTECTED USING INTUMESCENT PAINT.

14

ALL STRUCTURAL STEEL CONTRACTORS ARE TO BE CE CERTIFIED IN ACCORDANCE WITH EN 1090 & ISO 3834 TO EXC 2 WITH FULL TIME RESPONSIBLE WELDING COORDINATOR EXPERIENCED AND QUALIFIED TO TECHNICAL KNOWLEDGE LEVEL B OR BETTER TO WORKSHOP WITH SUITABLY QUALIFIED WELDERS TO FABRICATE IN ACCORDANCE WITH DRAWINGS AND SPECS. ALL NECESSARY WELDING PROCEDURE SPEC TO BE IN PLACE WITH QUALITY PLAN AND ALL STEEL SHOULD BE SUPPLIED IN ACCORDANCE WITH THE STRUCTURAL STEEL SPECIFICATION AND THE NATIONAL STRUCTURAL STEELWORK SPECIFICATION FOR BUILDING CONSTRUCTION (5TH EDITION)

15

HANDOVER PACK: MATERIALS CERTS TO BE PROVIDED WITH REGARD TO STEEL MATERIALS, (STEEL INSPECTION CERTIFICATES) WELDING WIRE, WELDING GAS, BOLTS AND PAINT INSPECTION CERTS. FACTORY PRODUCTION CONTROL CERTS TO BE PROVIDED TOGETHER WITH WELDING CERTS.

16

VISUAL INSPECTION TO 100% OF WELDS BY QUALIFIED WELD INSPECTOR. SUPPLEMENTARY NON-DSTRUCTIVE WELD TESTING IN ACCORDANCE WITH EN 1090 AND N.S.S.S.

1

CONCRETE CONSTRUCTION SHALL BE IN ACCORDANCE WITH IS EN 1992. CEMENT SHALL BE ORDINARY PORTLAND CEMENT COMPLYING WITH EN197-1 UNLESS OTHERWISE APPROVED. REINFORCEMENT SHALL BE HOT ROLLED DEFORMED BARS COMPLYING WITH EN10080. ALL HIGH YIELD BARS SHALL BE DEFORMED BARS TYPE 2. STEEL FABRIC SHALL COMPLY WITH BS 4483.

2

ALL CONCRETE SHALL BE IN ACCORDANCE WITH IS EN 206-2013.

3

NOMINAL CONCRETE COVER TO REINFORCEMENT SHALL BE AS FOLLOWS UNLESS OTHERWISE NOTED ON DRAWINGS :

*SURFACE POURED AGAINST EARTH, 75mm
*SURFACE POURED AGAINST FORMWORK 50mm
*BUT IN CONTACT WITH EARTH, 35mm
*BEAMS, COLUMNS, 35mm
*SLABS, 35mm

4

CONCRETE BLINDING SHALL BE C15. ALL OTHER CONCRETE TO BE C35N20

5

FOR CONCRETING IN COLD WEATHER REFER TO THE CONCRETE SPECIFICATION AND TO THE PROVISIONS OF EN 1992.

6

THE POSITION AND TREATMENT OF CONSTRUCTION JOINTS SHALL BE TO THE APPROVAL OF THE ENGINEER.

7

MOVEMENT JOINT FILLER SHALL BE A COMPRESSIBLE FIBRE BOARD SUCH AS HYDROCELL XL BY FOSROC INTERNATIONAL OR SIMILAR APPROVED.

8

MOVEMENT JOINT SEALANT SHALL BE A TWO PART POLYSULPHIDE SEALANT SUCH AS THIOFLEX 600 BY FOSROC INTERNATIONAL OR AN APPROVED EQUAL, UNLESS NOTED OTHERWISE.

9

CONCRETE VERTICAL SURFACES ADJOINING MASONRY WALLS SHALL BE CAST WITH A CONTINUOUS STAINLESS STEEL PROPRIETARY CHANNEL BY ANCON CCL OR SIMILAR APPROVED.

10

PROVIDE A MINIMUM OF 24 HOURS NOTICE TO THE ENGINEER FOR INSPECTION OF ALL REINFORCING STEEL PRIOR TO PLACING CONCRETE.

11

CONCRETE CUBES SHALL BE TAKEN AT A RATE OF ONE SET OF THREE CUBES PER 20m³ OR FRACTION THEREOF, WITH A MINIMUM OF ONE SET PER DAY PER CLASS OF CONCRETE. CUBES SHALL BE TESTED AT 7 AND 28 DAYS. ALL THE RESULTS SHALL BE FORWARDED DIRECTLY TO THE ENGINEER

12

ALL SAMPLING AND TESTING OF CONCRETE TO COMPLY WITH EN13791 & EN1290.

13

CORE DRILLING OF ANY REINFORCED CONCRETE ELEMENTS OF THE WORK SHALL NOT BE ALLOWED WITHOUT THE PRIOR WRITTEN APPROVAL OF THE ENGINEER.

14

100mm KICKERS FOR ALL RISING ELEMENTS.

15

SR21 FILL BELOW FLOOR SLABS WITH LOW SULPHUR / SULPHATE CONTENT.

PRECAST CONCRETE

1

CONCRETE CONSTRUCTION SHALL BE IN ACCORDANCE WITH IS EN 1992. CEMENT SHALL BE ORDINARY PORTLAND CEMENT COMPLYING WITH IS1 OR BS12 UNLESS OTHERWISE APPROVED. REINFORCEMENT SHALL BE HOT ROLLED DEFORMED BARS COMPLYING WITH IE 10080. ALL HIGH YIELD BARS SHALL BE DEFORMED BARS TYPE 2. STEEL FABRIC SHALL COMPLY WITH BS 4483.

2

ALL CONCRETE SHALL BE IN ACCORDANCE WITH IS EN 206-2013.

3

UNITS ARE TO BE ERECTED BY COMPETENT CONTRACTORS AS APPROVED BY PRECAST SUPPLIER, MANUFACTURER, AND THE ENGINEER.

4

OPENINGS - NO OPES, NOTCHES OR ALTERATIONS ARE TO BE MADE TO UNITS WITHOUT PRIOR CONSULTATION WITH PRECAST SUPPLIER/MANUFACTURER.

5

MORTAR FOR BEDDING SHALL COMPLY WITH THE MIX DESIGNATION FOR COMPRESSIVE STRENGTH CLASS M4 GIVEN IN TABLE 2 OF NATIONAL ANNEX TO IS EN 1996 UNLESS OTHERWISE STATED ON DRAWINGS. MORTAR TESTING SHALL BE IN ACCORDANCE WITH IS EN 1015-12. APPENDIX A.1. SAND FOR MORTAR SHALL COMPLY WITH BS 1199 AND BS1200.

6

THE CONTRACTOR SHALL SUBMIT FULLY DETAILED FABRICATION DRAWINGS TO THE ENGINEER FOR APPROVAL A MINIMUM OF 10 WORKING DAYS BEFORE FABRICATION IS DUE TO COMMENCE. NO FABRICATION SHALL COMMENCE UNTIL APPROVAL OF THE SHOP DRAWINGS IS RECEIVED AND UNTIL ALL COMMENTS HAVE BEEN INCORPORATED.

7

APPROVAL BY THE ENGINEER IN NO WAY RELIEVES THE CONTRACTOR FOR ANY RESPONSIBILITY FOR THE ACCURACY, CORRECTNESS AND ADEQUACY OF CALCULATIONS, DESIGN, DETAILS AND DIMENSIONS.

8

PROVIDE A MINIMUM OF 24 HOURS NOTICE TO THE ENGINEER FOR INSPECTION OF ALL REINFORCING STEEL PRIOR TO PLACING CONCRETE.

9

CONCRETE CUBES SHALL BE TAKEN AT A RATE OF ONE SET OF THREE CUBES PER 20m³ OR FRACTION THEREOF, WITH A MINIMUM OF ONE SET PER DAY PER CLASS OF CONCRETE. CUBES SHALL BE TESTED AT 7 AND 28 DAYS. ALL THE RESULTS SHALL BE FORWARDED DIRECTLY TO THE ENGINEER.

10

ALL SAMPLING AND TESTING OF CONCRETE TO COMPLY WITH BS 1881.

1

MASONRY CONSTRUCTION SHALL CONFORM WITH IS EN 771 AND IS EN 1996. BLOCKS SHALL BE CONCRETE MASONRY UNITS. CONFORMING TO IS EN 771. SAND FOR MORTAR SHALL COMPLY WITH EN 998-2. CEMENT SHALL BE ORDINARY PORTLAND CEMENT TO EN197-1.

2

BLOCKS FOR RISING WALLS SHALL BE SOLID UNITS. BLOCK SIZES FOR OTHER AREAS SHALL BE AS INDICATED ON DRAWINGS OR SPECIFICATIONS. JOINT THICKNESS SHALL BE 10mm.

3

THE CHARACTERISTIC COMPRESSIVE STRENGTH OF THE BLOCKS SHALL BE 7.5 N/mm2 UNLESS NOTED OTHERWISE ON DRAWINGS. BLOCKS SHALL BE TESTED AT A RATE OF 5 PER 1000 BY AN INDEPENDENT TESTING AGENCY.

4

GENERAL BLOCKWORK : MORTAR SHALL COMPLY WITH THE MIX DESIGNATED FOR COMPRESSIVE STRENGTH CLASS M4 GIVEN IN TABLE 2 OF NATIONAL ANNEX TO IS EN 1996 UNLESS OTHERWISE STATED ON DRAWINGS. MORTAR TESTING SHALL BE IN ACCORDANCE WITH IS EN1015-12.

5

RISING WALL BLOCKWORK : MORTAR SHALL COMPLY WITH THE MIX DESIGNATED FOR COMPRESSIVE STRENGTH CLASS M6 GIVEN IN TABLE 2 OF NATIONAL ANNEX TO IS EN 1996 UNLESS OTHERWISE STATED ON DRAWINGS. MORTAR TESTING SHALL BE IN ACCORDANCE WITH IS EN1015-12.

6

TYPICAL MOVEMENT JOINT SPACING FOR UNREINFORCED CONCRETE MASONRY PANELS INTERNALLY ARE BETWEEN 8m - 12m AND EXTERNALLY 7m MAX. IDEALLY PANELS SHOULD BE SQUARE, BUT GENERALLY THE LENGTH SHOULD NOT EXCEED 3 * HEIGHT OF THE PANEL.

7

RISING WALL INTERSECTIONS SHALL BE TIED TOGETHER IN A MASONRY BOND. OTHER WALLS SHALL BE TIED IN MASONRY BOND AT CORNERS UNLESS NOTED OTHERWISE ON DRAWINGS. ALL BLOCKWORK ADJACENT TO CONCRETE WALLS TO BE TIED TO CONCRETE USING WALL TIES COMPLYING WITH POINT 7.

8

CAVITY WALL TIES SHALL BE STAINLESS STEEL TYPE 1 TIES IN ACCORDANCE WITH EN 845-1 UNLESS OTHERWISE NOTED ON DRAWINGS. TIES SHALL BE SPACED IN ACCORDANCE WITH BS8103 AND PP6697.

9

FOR MASONRY CONSTRUCTION DURING COLD WEATHER REFER TO THE SPECIFICATIONS AND THE PROVISIONS OF IS EN 1996.

10

ALL RISING WALLS SHALL BE LAID WITH FULL MORTAR COVERAGE ON HORIZONTAL AND VERTICAL FACES.

11

THE CONTRACTOR SHALL BE RESPONSIBLE FOR SHORING AND BRACING OF ALL MASONRY WALLS AS NECESSARY TO ENSURE STABILITY DURING CONSTRUCTION. SUCH SHORING AND BRACING SHALL BE MAINTAINED IN POSITION UNTIL THE MASONRY HAS ATTAINED ITS DESIGN STRENGTH AND THE RESTRAINING SUPPORT STRUCTURE IS IN PLACE.

12

BACK FILL SHALL NOT BE PLACED AGAINST WALLS WITHIN 10 DAYS OF COMPLETION OF THE WALL. THE MAXIMUM LEVEL DIFFERENCE BETWEEN BACK FILL ON EITHER SIDE OF A MASONRY WALL SHALL BE 225mm. HEAVY EQUIPMENT USED IN BACK FILLING SHALL NOT OPERATE CLOSER TO THE WALL THAT A DISTANCE EQUAL TO THE HEIGHT OF THE BACK FILL ABOVE THE FOOTING.

13

REQUIREMENTS FOR REINFORCED MASONRY:
*CONCRETE FOR FILLING CORES OR EMBEDDING REINFORCING BARS SHALL BE 35N10 UNLESS NOTED OTHERWISE ON DRAWINGS. *MAXIMUM AGGREGATE SIZE 10mm. *VERTICAL BAR REINFORCEMENT SHALL BE SECURED IN POSITION *CAVITIES CONTAINING REINFORCEMENT SHALL BE COMPLETELY FILLED WITH CONCRETE AND CLEANED OUT OPENINGS SHALL BE PROVIDED AT THE BASE OF VERTICAL CORES.

14

FOR LINTEL DETAILS REFER TO PROJECT DETAIL DRAWINGS AND MASONRY SPECIFICATION.

15

PLASTER / RENDER SAND TO BE LOW IN IRON AND SULPHUR / SULPHATE CONTENT. POTABLE WATER ONLY WITH LOW IRON AND SULPHATE CONTENT TO BE USED IN RENDER / PLASTER MIX.

CHASES IN MASONRY WALLS

1

CHASES SHOULD NOT IMPAIR THE STABILITY OF THE WALL. CHASES ARE NOT ALLOWED WHEN USING HOLLOW BLOCKS

2

VERTICAL CHASES FORMED AFTER CONSTRUCTION OF MASONRY SHOULD NOT BE DEEPER THAN 30mm

3

THE CUMULATIVE WIDTH OF VERTICAL CHASES AND RECESSES SHOULD NOT EXCEED 0.13 TIMES THE LENGTH OF THE WALL

4

HORIZONTAL DISTANCE BETWEEN ADJACENT CHASES OR BETWEEN A CHASE AND A RECESS OR AN OPENING SHOULD NOT BE LESS THAN 225mm

5

VERTICAL AND HORIZONTAL OR INCLINED CHASES SHOULD COMPLY WITH I.S. EN 1996-1-1 TABLES FOR SIZES OF CHASES AND RECESS IN MASONRY

LOADINGS

1

ALL LOADINGS ARE IN ACCORDANCE WITH THOSE SPECIFIED IN IS EN1991

2

IMPOSED LOADINGS (EXCLUDING FINISHES AND SERVICES)

LOCATION

LOADING

POINT LOAD

LINTELS - PRECAST COMPOSITE

1

150mm BEARING UP TO 1.5m SPAN
200mm BEARING OVER 1.5m SPAN

2

LINTELS TO BE INSTALLED WITH ROUGH SIDE UP TO ACHIEVE MORTAR BOND WITH COMPOSITE BLOCKWORK ABOVE.

3

PROPS TO SPANS OVER 1.2m. MAX SPACING OF PROPS 1.2m. PROPS TO REMAIN IN POSITION UNTIL CONSTRUCTION HAS MATURED.

1

THE CONTRACTOR SHALL CALL FOR THE SUPERVISING ENGINEER'S INSPECTION AND OBTAIN HIS APPROVAL FOR WORK DONE BEFORE PROCEEDING FURTHER

2

WHEN THE SUBGRADE HAS BEEN BUILT UP TO FINAL LEVELS.

3

WHEN THE REINFORCEMENT HAS BEEN FIXED, CHECKED AND IS READY FOR CONCRETING.

4

BEFORE TAKING REMEDIAL ACTION TO CORRECT IMPERFECTIONS IN CONCRETE.

5

BEFORE FINISH PAINTING STEEL FRAMEWORKS.

6

BEFORE GROUTING-IN STEEL FRAMEWORKS.

7

BEFORE SHEETING OR BRICKING IN STEEL FRAMEWORKS.

CONCRETE NOTES:

1.

READ IN CONJUNCTION WITH CONCRETE SPECIFICATION

2.

ALL CONCRETE TO BE GRADE C28/35 UNLESS NOTED OTHERWISE

3.

SEE DRG. S001 FOR COVER TO REINFORCEMENT

4.

USE 75MM LEAN MIX CONCRETE BLINDING ON COMPACTED FORMATION TO ALL FOUNDATIONS

5.

ALL REINFORCING STEEL TO BE HIGH YIELD 460-500N/SQ.MM

6.

ALL CONCRETE TO BE FULLY VIBRATED TO ENSURE NO HONEYCOMBING

7.

ALL FORMWORK TO BE MADE WITH SHUTTERING PLYWOOD PROPPED AND BRACED AS NECESSARY

GENERAL BAR NOTATION:

T

=

TOP

T1

=

TOP LAYER OF TOP REINFORCEMENT

T2

=

BOTTOM LAYER OF TOP REINFORCEMENT

B

=

BOTTOM

B1

=

BOTTOM LAYER OF BOTTOM REINFORCEMENT

B2

=

TOP LAYER OF BOTTOM REINFORCEMENT

UB

=

UBARS

EF

=

EACH FACE

N1

=

OUTER LAYER OF NEAR FACE

N2

=

INNER LAYER OF NEAR FACE

F1

=

OUTER LAYER OF FAR FACE

F2

=

INNER LAYER OF FAR FACE

ABR

=

ALTERNATE BARS REVERSED

MINIMUM LAPS:

B10 - 500mm

B12 - 600mm

B16 - 800mm

B20 - 1000mm

B25 - 1250mm

LAP DIMENSIONS APPLY TO THE SMALLER OF THE LAPPING BARS.

TYPICAL REBAR CALL UP

10 B16 01-100 T

NOTES

SPACING OF BAR

BAR MARK REFERENCE

DIA OF BAR

GRADE OF STEEL

QUANTITY OF BAR

REINFORCING BARS WHICH EXCEED THE LENGTH GIVEN BELOW WEIGH MORE THAN 20KG. THE CONTRACTOR SHALL TAKE PRECAUTIONS TO AVOID THE RISK OF MANUAL HANDLING INJURIES. H20 = 8.1m H25 = 5.1m H32 = 3.2m H40 = 2.0m